# Counting Grids

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2210 | Mathematics | https://cses.fi/problemset/task/2210

Your task is to count the number of different $n \times n$ grids whose each square is black or white.

Two grids are considered to be different if it is not possible to rotate one of them so that they look the same.

## Input

The only input line has an integer $n$: the size of the grid.

## Output

Print one integer: the number of grids modulo $10^9+7$.

## Constraints

- $1 \le n \le 10^9$

## Example

| Input |
| --- |
| 4 |

| Output |
| --- |
| 16456 |